

# **Exhibit CC-93**

# DDR5 Module Part Numbering System

The part numbering system is available at [www.micron.com/numbering](http://www.micron.com/numbering)

MT

C

40

F

2

0

4

6

S

1

R

C

48B

A

1

T

MT = Micron Technology

Product Family  
C = DDR5 SDRAM

Number of DRAM Die

Voltage  
F = 12V (RDIMM)  
C = 5V (SODIMM, SOEDIMM, CSODIMM, UDIMM, EUDIMM, CUDIMM)

Package Ranks  
1 Rank  
2 Ranks  
3 Ranks  
4 Ranks  
8 Ranks

Logical Ranks (3DS or WB)  
0 = None  
2 = Total Logical Ranks  
4 = Total Logical Ranks  
8 = Total Logical Ranks  
16 = Total Logical Ranks

Component Config  
4 = x4  
8 = x8  
16 = x16

Module Density  
3 = 8GB (2<sup>33</sup>)  
Z = 12GB (2<sup>33.5</sup>)  
4 = 16GB (2<sup>34</sup>)  
Y = 24GB (2<sup>34.5</sup>)  
5 = 32GB (2<sup>35</sup>)  
X = 48GB (2<sup>35.5</sup>)  
6 = 64GB (2<sup>36</sup>)  
W = 96GB (2<sup>36.5</sup>)  
7 = 128GB (2<sup>37</sup>)  
V = 192GB (2<sup>37.5</sup>)  
8 = 256GB (2<sup>38</sup>)  
9 = 512GB (2<sup>39</sup>)  
A = 1024GB (2<sup>40</sup>)  
B = 2048GB (2<sup>41</sup>)  
C = 4096GB (2<sup>42</sup>)

Module Height  
S = LP  
D = VLP  
H = 2U  
L = 4U

Die-in-Package  
1 = Single-Die  
2 = Dual-Die  
4 = Quad-Die  
8 = Octal-Die

Final Pack Type - Optional - for  
CPG Distribution Parts Only

Mark

Shipment Packaging

R

Retail

T

Tray

Printed Circuit Board Revision  
Numeric = Production  
Alpha = Engineering Sample

Die Revision (Part Mark Designator)

DDR5 SDRAM Module Speed

| Module Speed Bin (Part Mark) | Component JEDEC Speed Bin | Component Speed Grade | Module Clock Frequency (MHz) | Module Data Rate (MTPS) | CL-nRCD-nRP |
|------------------------------|---------------------------|-----------------------|------------------------------|-------------------------|-------------|
| 32B                          | DDR5-3200B                | PC5-3200              | 1600                         | 3200                    | 26-26-26    |
| 36B                          | DDR5-3600B                | PC5-3600              | 1800                         | 3600                    | 30-30-30    |
| 40B                          | DDR5-4000B                | PC5-4000              | 2000                         | 4000                    | 32-32-32    |
| 44B                          | DDR5-4400B                | PC5-4400              | 2200                         | 4400                    | 36-36-36    |
| 48B                          | DDR5-4800B                | PC5-4800              | 2400                         | 4800                    | 40-39-39    |
| 52B                          | DDR5-5200B                | PC5-5200              | 2600                         | 5200                    | 42-42-42    |
| 56B                          | DDR5-5600B                | PC5-5600              | 2800                         | 5600                    | 46-45-45    |
| 60B                          | DDR5-6000B                | PC5-6000              | 3000                         | 6000                    | 48-48-48    |
| 64B                          | DDR5-6400B                | PC5-6400              | 3200                         | 6400                    | 52-52-52    |
| 68B                          | DDR5-6800B                | PC5-6800              | 3400                         | 6800                    | 56-56-56    |
| 72B                          | DDR5-7200B                | PC5-7200              | 3600                         | 7200                    | 58-58-58    |
| 76B                          | DDR5-7600B                | PC5-7600              | 3800                         | 7600                    | 62-61-61    |
| 80B                          | DDR5-8000B                | PC5-8000              | 4000                         | 8000                    | 64-64-64    |
| 88B                          | DDR5-8800B                | PC5-8800              | 4400                         | 8800                    | 72-71-71    |

Package Temperature Capability

| Code | Definition      |
|------|-----------------|
| C    | Commercial temp |
| I    | Industrial temp |
| B    | Extended temp   |

Module Type

| CODE | DESCRIPTION               |
|------|---------------------------|
| S    | 262-pin SODIMM X64        |
| V    | 262-pin CSODIMM X64       |
| T*   | 262-pin SOEDIMM X72       |
| W*   | 262-pin CSODIMM X72       |
| U    | 288-pin UDIMM X64         |
| A    | 287-pin CUDIMM X64        |
| E*   | 288-pin EUDIMM X72        |
| B*   | 287-pin CUDIMM X72        |
| R    | 288-pin/287-pin RDIMM X80 |
| P    | 288-pin/287-pin RDIMM X72 |

\* Die count increment of 10 (not 9)



# DDR4 Module Part Numbering System

The part numbering system is available at [www.micron.com/numbering](http://www.micron.com/numbering)

## DDR4 SDRAM modules

MT

A

36

A

SF

2G

72

P

Z

-

2G3

A

1

T

MT = Micron Technology

CC = Crucial & Components (Internal only)

Product Family

A = DDR4 SDRAM

Number of Die

Voltage

A = 1.2V

Module Options

TF = FBGA w/out Temp Sensor

TS = Dual-Die w/out Temp Sensor

TQ = Quad-Die w/out Temp Sensor

SF = FBGA w/Temp Sensor

SS = Dual-Die w/Temp Sensor

SQ = Quad-Die w/Temp Sensor

SE = Octal-Die w/Temp Sensor

DF = VLP w/Temp Sensor

DS = VLP Dual-Die w/Temp Sensor

DQ = VLP Quad-Die w/Temp Sensor

LF = 4U height with temp sensor

LS = 4U height Dual-Die with temp sensor

LQ = 4U height Quad-Die with temp sensor

LE = 4U height Octal-Die with temp sensor

HF = 2U height with temp sensor

HS = 2U height Dual-Die with temp sensor

HQ = 2U height Quad-Die with temp sensor

HE = 2U height Octal-Die with temp sensor

SZF = FBGA w/Temp Sensor and Heat Spreader

SZS = Dual-Die w/Temp Sensor and Heat Spreader

SZQ = Quad-Die w/Temp Sensor and Heat Spreader

SZE = Octal-Die w/Temp Sensor and Heat Spreader

DZF = VLP w/Temp Sensor and Heat Spreader

DZS = VLP Dual-Die w/Temp Sensor and Heat Spreader

DZQ = VLP Quad-Die w/Temp Sensor and Heat Spreader

Module Configuration

Depth, Width

Blank = Megabits

G = Gigabits

Final Pack Type - Optional - for CPG Distribution Parts Only

Mark

Shipment Packaging

R

Retail

T

Tray

Printed Circuit Board Revision

Die Revision (up to 2 characters)

DDR4 SDRAM Module Speed

| Module Speed Grade | Component Speed Grade/ Part Mark | Component Speed Bin | Clock Freq. (MHz) | Data Rate (MT/s) | Module Bandwidth | Module Configuration (CL-nRCD-nRP) |
|--------------------|----------------------------------|---------------------|-------------------|------------------|------------------|------------------------------------|
| -2G1               | -093E                            | DDR4-2133           | 1067              | 2133             | PC4-2133         | 15-15-15                           |
| -2S1               | -093H*                           | DDR4-2133           | 1067              | 2133             | PC4-2133         | 18-15-15                           |
| -2G3               | -083                             | DDR4-2400           | 1200              | 2400             | PC4-2400         | 17-17-17                           |
| -2G4               | -083E                            | DDR4-2400           | 1200              | 2400             | PC4-2400         | 16-16-16                           |
| -2S3               | -083H*                           | DDR4-2400           | 1200              | 2400             | PC4-2400         | 20-18-18                           |
| -2S4               | -083J*                           | DDR4-2400           | 1200              | 2400             | PC4-2400         | 19-17-17                           |
| -2G6               | -075                             | DDR4-2666           | 1333              | 2666             | PC4-2666         | 19-19-19                           |
| -2S6               | -075H*                           | DDR4-2666           | 1333              | 2666             | PC4-2666         | 22-19-19                           |
| -2G7               | -075E                            | DDR4-2666           | 1333              | 2666             | PC4-2666         | 18-18-18                           |
| -2G9               | -068                             | DDR4-2933           | 1467              | 2933             | PC4-2933         | 21-21-21                           |
| -2S9               | -068H*                           | DDR4-2933           | 1467              | 2933             | PC4-2933         | 24-21-21                           |
| -3G2               | -062E                            | DDR4-3200           | 1600              | 3200             | PC4-3200         | 22-22-22                           |
| -3S2               | -062H*                           | DDR4-3200           | 1600              | 3200             | PC4-3200         | 26-22-22                           |

SPD = serial presence-detect pin (module only)

\* Master Slave (MS)

Package Codes

| Pb-Free Devices | Package Descriptions                                                |
|-----------------|---------------------------------------------------------------------|
| Z               | Commercial temp; Green; 1Rx8, 1Rx4, 2Rx4, 4Rx4, 8Rx4 RDIMM & LRDIMM |
| DZ              | Commercial temp; Green; 2Rx8, 4Rx8, 8Rx8 RDIMM                      |
| IZ              | Industrial temp; Green; 1Rx8, 1Rx4, 2Rx4, 4Rx4, 8Rx4 RDIMM & LRDIMM |
| BZ              | Extended temp; Green; 1Rx8, 1Rx4, 2Rx4, 4Rx4, 8Rx4 RDIMM & LRDIMM   |
| DBZ             | Extended temp; Green; 2Rx8, 4Rx8, 8Rx8 RDIMM                        |

Module Type

|                                     | TAB_COUNT | FORM_FACTOR | BUFFERED |
|-------------------------------------|-----------|-------------|----------|
| A = 288-pin UDIMM (unbuffered)      | 288       | UDIMM       | FALSE    |
| H = 260-pin SODIMM                  | 260       | SODIMM      | FALSE    |
| L = 288-pin LRDIMM                  | 288       | LRDIMM      | TRUE     |
| LS = 288-pin 3DS (M/S) LRDIMM       | 288       | LRDIMM      | TRUE     |
| P = 288-pin RDIMM                   | 288       | RDIMM       | TRUE     |
| PS = 288-pin 3DS (M/S) RDIMM        | 288       | RDIMM       | TRUE     |
| AK = 288-pin miniUDIMM (unbuffered) | 288       | MINIUDIMM   | FALSE    |
| PK = 288-pin miniRDIMM              | 288       | MINIRDIMM   | TRUE     |
| N = 84-pin DDIMM                    | 84        | DDIMM       | TRUE     |
| X = 84-pin DDIMM (w/out registers)  | 84        | DDIMM       | FALSE    |

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# DDR3 Module Part Numbering System

The part numbering system is available at [www.micron.com/numbering](http://www.micron.com/numbering)

## DDR3 SDRAM modules

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|----------|-----------|-----------|-----------|----------|----------|----------|------------|----------|----------|---------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>MT</b> | <b>36</b> | <b>K</b> | <b>SF</b> | <b>2G</b> | <b>72</b> | <b>P</b> | <b>Z</b> | <b>-</b> | <b>1G6</b> | <b>E</b> | <b>1</b> |                                       |
| <b>Micron Technology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |           |          |           |           |           |          |          |          |            |          |          | <b>MB Vendor/Revision*</b>            |
| <b>Number of Die</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |           |          |           |           |           |          |          |          |            |          |          | <b>Printed Circuit Board Revision</b> |
| <b>Voltage</b><br>J = 1.5V<br>K = 1.35V<br>M = 1.35V reduced standby                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |           |          |           |           |           |          |          |          |            |          |          | <b>Die Revision</b>                   |
| <b>Module Options</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
| TF = FBGA<br>TS = DDP (dual die in package)<br>SF = FBGA with temp sensor<br>SS = DDP with temp sensor<br>BF = VLP (17.9mm) with temp sensor<br>BS = VLP (17.9mm) DDP with temp sensor<br>DF = VLP (18.75mm) with temp sensor<br>DS = VLP (18.75mm) DDP with temp sensor<br>GF = 1.5U height FBGA with temp sensor<br>GS = 1.5U height DDP with temp sensor<br>HF = 2U height with temp sensor<br>LF = 4U height with temp sensor<br>SZF = FBGA with temp sensor and heat spreader<br>SZS = DDP with temp sensor and heat spreader<br>SZQ = QDP with temp sensor and heat spreader<br>BZF = VLP (17.9mm) with temp sensor and heat spreader<br>BZS = VLP (17.9mm) DDP with temp sensor and heat spreader<br>DYS = VLP (18.75mm) DDP with temp sensor and alternate heat spreader<br>DZF = VLP (18.75mm) with temp sensor and heat spreader<br>DZS = VLP (18.75mm) DDP with temp sensor and heat spreader<br>DZQ = VLP (18.75mm) QDP with temp sensor and heat spreader |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
| <b>Module Configuration</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
| Depth, Width<br>Blank = Megabits<br>G = Gigabits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
| <b>Module Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |           |          |           |           |           |          |          |          |            |          |          |                                       |
| Blank = 240-pin registered DIMM<br>A = 240-pin unbuffered DIMM<br>H = 204-pin SODIMM<br>HS = 204-pin SODIMM (R/C G: standard layout)<br>HR = 204-pin SODIMM (R/C H: reverse layout)<br>P = 240-pin parity RDIMM<br>L = 240-pin LRDIMM<br>AK = 244-pin unbuffered miniDIMM<br>PK = 244-pin parity miniRDIMM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |           |          |           |           |           |          |          |          |            |          |          |                                       |

| DDR3 SDRAM Module Speed |                                 |                             |                   |                  |                  |                                        |
|-------------------------|---------------------------------|-----------------------------|-------------------|------------------|------------------|----------------------------------------|
| Module Speed Grade      | Component Speed Grade/Part Mark | JEDEC Component Speed Grade | Clock Freq. (MHz) | Data Rate (MT/s) | Module Bandwidth | Module Configuration SPD (CL-nRCD-nRP) |
| -80B                    | -25                             | DDR3-800                    | 400               | 800              | PC3-6400         | 6-6-6                                  |
| -80C                    | -25E                            | DDR3-800                    | 400               | 800              | PC3-6400         | 5-5-5                                  |
| -1G0                    | -187                            | DDR3-1066                   | 533               | 1066             | PC3-8500         | 8-8-8                                  |
| -1G1                    | -187E                           | DDR3-1066                   | 533               | 1066             | PC3-8500         | 7-7-7                                  |
| -1G2                    | -187F                           | DDR3-1066                   | 533               | 1066             | PC3-8500         | 6-6-6                                  |
| -1G3                    | -15                             | DDR3-1333                   | 667               | 1333             | PC3-10600        | 10-10-10                               |
| -1G4                    | -15E                            | DDR3-1333                   | 667               | 1333             | PC3-10600        | 9-9-9                                  |
| -1G5                    | -15F                            | DDR3-1333                   | 667               | 1333             | PC3-10600        | 8-8-8                                  |
| -1GA**                  | -125                            | DDR3-1333                   | 667               | 1333             | PC3-10600        | 11-11-11                               |
| -1G6                    | -125                            | DDR3-1600                   | 800               | 1600             | PC3-12800        | 11-11-11                               |
| -1G7                    | -125E                           | DDR3-1600                   | 800               | 1600             | PC3-12800        | 10-10-10                               |
| -1G8                    | -125F                           | DDR3-1600                   | 800               | 1600             | PC3-12800        | 9-9-9                                  |
| -1G9                    | -107                            | DDR3-1866                   | 933               | 1866             | PC3-14900        | 13-13-13                               |
| -1GB                    | -107E                           | DDR3-1866                   | 933               | 1866             | PC3-14900        | 12-12-12                               |
| -1GC                    | -107F                           | DDR3-1866                   | 933               | 1866             | PC3-14900        | 11-11-11                               |
| -2G1                    | -093                            | DDR3-2133                   | 1067              | 2133             | PC3-17000        | 14-14-14                               |
| -2G2                    | -093E                           | DDR3-2133                   | 1067              | 2133             | PC3-17000        | 13-13-13                               |
| -2G3                    | -093F                           | DDR3-2133                   | 1067              | 2133             | PC3-17000        | 12-12-12                               |

\*\* -1GA = -1333 SPD with -125 tested DRAM  
SPD = serial presence-detect pin (module only)

| Package Codes   |                                                                                  |
|-----------------|----------------------------------------------------------------------------------|
| Pb-Free Devices | Package Descriptions                                                             |
| Y               | Commercial temp; single-, dual- or quad-rank DIMM                                |
| Z               | Halogen-free; commercial temp; single-, dual-, quad- or octal-rank DIMM          |
| IZ              | Halogen-free; Industrial temp, commercial temp; single-, dual- or quad-rank DIMM |
| TZ              | Halogen-free; Industrial temp; select dual- and quad-rank DIMM                   |
| DY              | Commercial temp; select dual- or quad-rank DIMM                                  |
| DZ              | Halogen-free; commercial temp; select dual-, quad- or octal-rank DIMM            |

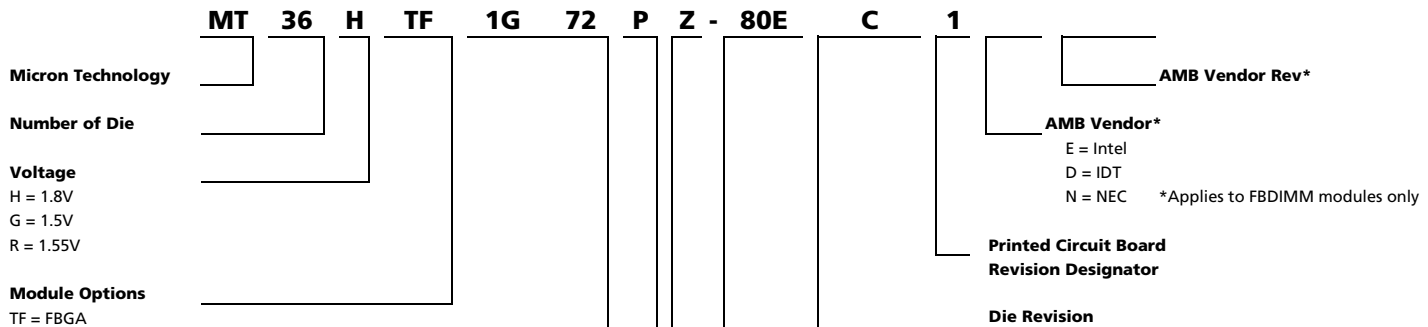
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The part numbering system is available at [\*\*www.micron.com/numbering\*\*](http://www.micron.com/numbering)

## DDR2 SDRAM modules



### DDR2 SDRAM Module Speed

| Module Speed Grade | Component Speed Grade/ Part Mark | JEDEC Component Speed Grade | Clock Frequency (MHz) | Data Rate (MT/s) | Module Bandwidth | Module Configuration SPD (CL~RCD~nRP) |
|--------------------|----------------------------------|-----------------------------|-----------------------|------------------|------------------|---------------------------------------|
| -40E               | -5E                              | DDR2-400                    | 200                   | 400              | PC2-3200         | 3-3-3                                 |
| -53E               | -37E                             | DDR2-533                    | 267                   | 533              | PC2-4200         | 4-4-4                                 |
| -667               | -3                               | DDR2-667                    | 333                   | 667              | PC2-5300         | 5-5-5                                 |
| -80E               | -25E                             | DDR2-800                    | 400                   | 800              | PC2-6400         | 5-5-5                                 |
| -800               | -25                              | DDR2-800                    | 400                   | 800              | PC2-6400         | 6-6-6                                 |
| -1GA               | -187E                            | DDR2-1066                   | 533                   | 1066             | PC2-8500         | 7-7-7                                 |

SPD = serial presence-detect pin (module only)

### Package Codes

| Pb-Free Devices | Package Descriptions                                            |
|-----------------|-----------------------------------------------------------------|
| Y               | Commercial temp; single- or dual-rank DIMM                      |
| Z               | Halogen-free; commercial temp; single- or dual-rank DIMM        |
| DY              | Commercial temp; select dual-rank                               |
| DZ              | Halogen-free; commercial temp; select dual- or quad-rank DIMM   |
| IY              | Industrial temp; select single and dual-rank DIMM               |
| TY              | Industrial temp; select dual and quad-rank DIMM                 |
| IZ              | Halogen-free; industrial temp; select single and dual-rank DIMM |
| TZ              | Halogen-free; industrial temp; select dual and quad-rank DIMM   |

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# DDR/SDRAM Module Part Numbering System

The part numbering system is available at [www.micron.com/numbering](http://www.micron.com/numbering)

## DDR/SDRAM modules

MT

36

V

DDF

256

72

Y

-

40B

J

2

Micron Technology

Number of Die

Voltage

L = 3.3V (SDRAM)

V = 2.5V (DDR SDRAM)

Module Options

DDF = DDR FBGA

DDT = DDR TSOP

DVF = DDR FBGA VLP

SDF = SDRAM FBGA

SDT = SDRAM TSOP

Module Configuration

Depth, Width

Blank = Megabits

G = Gigabits

Module Type\*

Blank = 168-pin/184-pin/240-pin registered DIMM

A = 168-pin/184-pin/240-pin unbuffered DIMM

H = 144-pin/200-pin SODIMM

PH = 144-pin/200-pin unbuffered SODIMM with PLLs

U = 100-pin unbuffered DIMM

\*All SDRAM and DDR DIMMs have serial-presence detect.

Printed Circuit Board

Revision Designator

Die Revision Designator

DDR Module Speed

| Module Speed Grade | Component Speed Grade/ Part Mark | JEDEC Component Speed Grade | Clock Frequency (MHz) | Data Rate (MT/s) | Module Bandwidth | Module Configuration SPD (CL-nRCD-nRP) |
|--------------------|----------------------------------|-----------------------------|-----------------------|------------------|------------------|----------------------------------------|
| -335               | -6T                              | DDR333 TSOP                 | 167                   | 333              | PC2700           | 2.5-3-3                                |
| -335               | -6                               | DDR333 FBGA                 | 167                   | 333              | PC2700           | 2.5-3-3**                              |
| -40B               | -5B                              | DDR400*                     | 200                   | 400              | PC3200           | 3-3-3                                  |

SPD = serial presence-detect pin (module only)

\*DDR400 nominal voltage is 2.6V

\*\*Data valid window is 150ps greater than -6T

Package Codes

| Lead-Plated Devices | Pb-Free Devices | Package Descriptions                       |
|---------------------|-----------------|--------------------------------------------|
| G                   | Y               | Commercial temp; single- or dual-rank DIMM |
| DG                  | DY              | Commercial temp; select dual-rank          |
| I                   | IY              | Industrial temp DIMM                       |
| T                   | TY              | Industrial temp; select dual-rank DIMM     |

| SDR Modules  |                           |                                       |
|--------------|---------------------------|---------------------------------------|
| Module Speed | Allowable Component Speed | CL <sup>-1</sup> RCD <sup>-1</sup> RP |
| -133         | -6A, -75, -7E             | 3-3-3                                 |
| -13E         | -7E                       | 2-2-2                                 |

| DDR Modules  |                           |
|--------------|---------------------------|
| Module Speed | Allowable Component Speed |
| -335         | -6, -6T, -5B              |
| -40B         | -5B                       |



# DDR4 NVDIMM Part Numbering System

The part numbering system is available at [www.micron.com/numbering](http://www.micron.com/numbering)

## DDR4 SDRAM Nonvolatile modules

MT

A

18

A

SF

2G

72

XF

1

Z

-

2G9

WP

1

AB

Micron Technology

Product Family

Number of DRAM Die

Voltage

Module Options

Module Configuration

Module Type

A = DDR4

SF = FBGA with temp sensor

SS = DDP with temp sensor

SQ = QDP with temp sensor

SZF = FBGA with temp sensor and heat spreader

SZS = DDP with temp sensor and heat spreader

SZQ = QDP with temp sensor and heat spreader

DZF = VLP (22mm) with temp sensor and heat spreader

Depth, Width

Blank = Megabits

G = Gigabits

DDR4: PF = 288-pin parity NAND Based NVRDIMM (Save-N, with Pullup Resistor)

DDR4: XF = 288-pin parity NAND Based NVRDIMM (Save-N, No Pullup Resistor)

DDR4: LF = 288-pin NAND Based NVLRDIMM

Controller Type (JEDEC compliant)

Printed Circuit Board Revision

Design ID Designator (Die Rev.)

DDR4/DDR3 SDRAM Module Speed

| DRAM Technology | Module Speed Grade | Component Speed Grade/Part Mark | JEDEC Component Speed Grade | Clock Freq. (MHz) | Data Rate (MT/s) | Module Bandwidth | Module Configuration SPD (CL-nRCD-nRP) |
|-----------------|--------------------|---------------------------------|-----------------------------|-------------------|------------------|------------------|----------------------------------------|
| DDR4 SDRAM      | -1G6               | -125E                           | DDR4-1600                   | 800               | 1600             | PC4-1600         | 11-11-11                               |
|                 | -1G9               | -107E                           | DDR4-1866                   | 933               | 1866             | PC4-1866         | 13-13-13                               |
|                 | -1S9               | -107H                           | DDR4-1866                   | 933               | 1866             | PC4-1866         | 14-13-13                               |
|                 | -2G1               | -093E                           | DDR4-2133                   | 1067              | 2133             | PC4-2133         | 15-15-15                               |
|                 | -2S1               | -093H                           | DDR4-2133                   | 1067              | 2133             | PC4-2133         | 16-15-15                               |
|                 | -2G3               | -083                            | DDR4-2400                   | 1200              | 2400             | PC4-2400         | 17-17-17                               |
|                 | -2G4               | -083E                           | DDR4-2400                   | 1200              | 2400             | PC4-2400         | 16-16-16                               |
|                 | -2G6               | -075                            | DDR4-2666                   | 1333              | 2666             | PC4-2666         | 19-19-19                               |
|                 | -2G9               | -068                            | DDR4-2933                   | 1467              | 2933             | PC4-2933         | 21-21-21                               |
|                 | -3G2               | -062E                           | DDR4-3200                   | 1600              | 3200             | PC4-3200         | 22-22-22                               |

SPD = serial presence-detect pin (module only)

Package Codes

| Pb-Free Devices | Package Descriptions                                                             |
|-----------------|----------------------------------------------------------------------------------|
| Z               | Halogen-free; commercial temp; single-, dual-, quad- or octal-rank DIMM          |
| DZ              | Halogen-free; commercial temp; select dual-, quad- or octal-rank DIMM            |
| IZ              | Halogen-free; Industrial temp, commercial temp; single-, dual- or quad-rank DIMM |
| TZ              | Halogen-free; Industrial temp; select dual- and quad-rank DIMM                   |

Flash Type

1 = Initial version of module

2-9 = Subsequent version of module with different Flash memory

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